Input Ontology (IO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *INJONGO* | ***Context:*** *Urban Ethiopian Utility Chatbot Users — Amharic-Language Telegram/SMS*

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

INJONGO includes a Utility domain explicitly selected as 'most suitable to African contexts' [Q39, Q117] and Amharic data does contain utility-adjacent bill_balance examples [D1, D8]. However, for the Ethiopian deployment specifically, the dimension is HIGH priority and the gaps are concrete and severe: prepaid meter top-up, STS token entry, outage reporting, load-shedding schedule queries, and account/meter lookup intents are not represented in the 19-example Amharic sample (D20) and are not enumerated in the documented intent list. The paper itself acknowledges domains 'essential for real-world applications' are missing [Q107] and notes the fixed uniform distribution may not reflect real-world frequencies [Q111]. Additionally, the bill_balance intent conflates utility billing with cable TV, retail credit, and phone bills across language splits (D1 vs D22), introducing construct-irrelevant variance for an EEU/AAWSA-scoped deployment. The 40-intent ceiling is also flagged as a source of measurement noise [Q86].

ID	Evidence
Q39	“The source data for our multilingual benchmark is from the CLINC English dataset—an intent detection with 150 intent classes across 10 domains (but without slot annotation), we extracted 40 intents from five most suitable domains to the African contexts: Banking (e.g. "transfer", "pay bill"), Home (e.g. "play music", "calendar update"), Kitchen and Dining (e.g. "recipe", "confirm reservation"), Travel (e.g. "exchange rate", "book flight"), and Utility (e.g.” (p.3)
Q86	“For intent detection, we find that all open models achieved below 50% on the relatively easy task of intent detection. The poor performance may be attributed to either a lack of exposure to many African languages or the large label space (i.e. 40) for the classification task.” (p.7)
Q107	“The scope of INJONGO is constrained by its coverage of only 5 domains and 40 intents, missing some other domains such as healthcare and education that are essential for real-world applications.” (p.9)
Q111	“Additionally, the fixed distribution of examples across intents may not accurately reflect the natural frequency of these interactions in real-world conversations.” (p.9)
Q117	“These are a total of 40 categories across 5 domains (Banking, Kitchen and Dining, Travel, Utility, and Home).” (p.13)
D1	"በወር ለመብራት ስንት ነው መክፈል የሚገባኝ?" — BILL_TYPE: ለመብራት — "How much do I need to pay for electricity per month?" — electricity utility bill query in Amharic
D8	"የዚህ ወር መከፈል ያለባቸውን የፍጆታ ዝርዝር አሳየኝ።" — DATE: የዚህ ወር — "Show me the list of utility expenses to be paid this month." — direct utility bill balance query
D20	"ወደ ቱርክ በፍጥነት የምደርስበት የበረራ አማራጭ ንገረኝ።" — COUNTRY: ቱርክ — "Tell me a flight option to Turkey quickly." — Travel domain; no utility-specific intents observed in 19 Amharic samples
D22	"Za ku iya tabbatar min ina da ajiyar tebur a taco bell da ƙarfe 7 na yamma" — RESTAURANT_NAME: taco bell — Hausa: "Can you confirm I have a table reservation at Taco Bell at 7 PM?" — Western fast food chain in culturally-specific dataset

Strengths

- Utility is one of five named domains [Q5, Q39] and the bill_balance intent contains genuine electricity utility content in Amharic (D1: ለመብራት annotated as BILL_TYPE)
- Cross-lingual evidence confirms electricity and water billing intents are operationally populated (D17 Luganda, D18 Kinyarwanda), so the utility category is not a stub
- Ground-up African design rather than translated CLINC reduces baseline Western bias in category selection [Q39, Q101]

ID	Evidence
Q5	“We cover the following five domains: banking, home, travel, utility, and kitchen & dining.” (p.1)
Q39	“The source data for our multilingual benchmark is from the CLINC English dataset—an intent detection with 150 intent classes across 10 domains (but without slot annotation), we extracted 40 intents from five most suitable domains to the African contexts: Banking (e.g. "transfer", "pay bill"), Home (e.g. "play music", "calendar update"), Kitchen and Dining (e.g. "recipe", "confirm reservation"), Travel (e.g. "exchange rate", "book flight"), and Utility (e.g.” (p.3)
Q101	“Figure 4 shows our final experiments that compare cross-lingual transfer results from two English datasets: CLINC (Western-centric) and INJONGO (African-centric) on the intent detection task.” (p.8)
D1	"በወር ለመብራት ስንት ነው መክፈል የሚገባኝ?" — BILL_TYPE: ለመብራት — "How much do I need to pay for electricity per month?" — electricity utility bill query in Amharic
D17	"Obukuumi n'amasannyalaze nteekeddwa kubisasula ddi?" — BILL_TYPE: Obukuumi, amasannyalaze — Luganda: "When do I need to pay the water and electricity bills?" — two BILL_TYPE slots; "amasannyalaze" = electricity
D18	"Mu bisanzwe nakwishyura fagitire y'amafaranga angahe ku mazi n'umuriro" — BILL_TYPE: mazi n'umuriro — Kinyarwanda: "How much do I normally pay for water and electricity bills?" — utility billing domain confirmed

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required deployment categories include prepaid meter top-up, STS token entry, outage report, outage status query, load-shedding schedule, bill dispute, payment history, account/meter lookup, payment via Telebirr/CBE Birr, water complaints, and human-agent escalation (per elicitation and YAML deployment_required_intents).

IO-2: Yes — confirmed omissions: no prepaid meter top-up, STS token, outage report, or load-shedding intents observed in 19 Amharic examples (D20) and not enumerated in documented intent list; paper explicitly notes 'healthcare and education' missing and 5/40 may miss real-world domains [Q107]. Flagged gap_id 1 in YAML resolved as NOT FOUND for utility-specific Ethiopian intents.

IO-3: Yes, partially — domains like Kitchen & Dining, Home (alarm, calendar_update, music), Travel (book_flight: D5, D20), and non-utility billing (cable TV, retail credit cards in bill_balance D22, D21) are present in the 40-intent space but irrelevant to an EEU/AAWSA chatbot, consuming evaluation weight when averaged over the 40-intent label space [Q86].

IO-4: Content underrepresentation gap: deployment-critical utility intents absent. Construct-irrelevant variance gap: heterogeneous bill_balance scope and non-utility domains inflate label space without serving deployment. Distributional mismatch: uniform 80-per-intent distribution [Q56] does not reflect skewed real-world utility interaction frequencies [Q111].

ID	Evidence
Q39	“The source data for our multilingual benchmark is from the CLINC English dataset—an intent detection with 150 intent classes across 10 domains (but without slot annotation), we extracted 40 intents from five most suitable domains to the African contexts: Banking (e.g. "transfer", "pay bill"), Home (e.g. "play music", "calendar update"), Kitchen and Dining (e.g. "recipe", "confirm reservation"), Travel (e.g. "exchange rate", "book flight"), and Utility (e.g.” (p.3)
Q56	“Our final annotation resulted in 3,200 annotated utterances, with 80 utterances per intent for each of the 16 African languages.” (p.5)
Q86	“For intent detection, we find that all open models achieved below 50% on the relatively easy task of intent detection. The poor performance may be attributed to either a lack of exposure to many African languages or the large label space (i.e. 40) for the classification task.” (p.7)

Q107	“The scope of INJONGO is constrained by its coverage of only 5 domains and 40 intents, missing some other domains such as healthcare and education that are essential for real-world applications.” (p.9)
Q111	“Additionally, the fixed distribution of examples across intents may not accurately reflect the natural frequency of these interactions in real-world conversations.” (p.9)
D5	"ከድሬዳዋ ወደ ባህርዳር ለመስከረም አምስት በረራ ያዝልኝ።" — CITY_OR_PROVINCE: ድሬዳዋ, ባህርዳር — "Book me a flight from Dire Dawa to Bahir Dar for the fifth of Meskerem." — Ethiopian cities and Ethiopian calendar month
D20	"ወደ ቱርክ በፍጥነት የምደርስበት የበረራ አማራጭ ንገረኝ።" — COUNTRY: ቱርክ — "Tell me a flight option to Turkey quickly." — Travel domain; no utility-specific intents observed in 19 Amharic samples
D21	"Ke hloka bokae ho lefa karete ya Edgars" — spans: [] — Sesotho: "How much do I need to pay the Edgars card?" — Edgars is South African retail credit card; no slots annotated despite implied BILL_TYPE/SHOPPING_ITEM
D22	"Za ku iya tabbatar min ina da ajiyar tebur a taco bell da karfe 7 na yamma" — RESTAURANT_NAME: taco bell — Hausa: "Can you confirm I have a table reservation at Taco Bell at 7 PM?" — Western fast food chain in culturally-specific dataset
WEB-11	gap_id 4 — no Ethiopian utility-specific NLP slot extraction literature exists; gap is documentation-confirmed (https://arxiv.org/pdf/2502.09814)

Information Gaps

- Full enumeration of the eight Utility-domain intent labels in INJONGO is not surfaced (gap_id 1)
- Full Amharic intent distribution beyond 19 sampled examples is unknown — utility-domain content density at scale is unquantified

Requires Expert Verification

- EEU/AAWSA production intent frequency distribution to compare against INJONGO's uniform 80-per-intent design
- Whether INJONGO's documented utility intents (e.g., 'pay bill') map onto EEU prepaid top-up workflow conventions

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Remediation

Gap 1: Deployment-critical utility intents (prepaid meter top-up, STS token entry, outage report, load-shedding schedule query, account/meter lookup) are absent from the documented 40-intent taxonomy.

Recommendation: Define an extended intent set with EEU and AAWSA customer service leads; collect 80–200 seed utterances per new intent in Amharic via supervised elicitation from Ethiopian native speakers familiar with the utility workflow; concatenate to INJONGO Amharic data for fine-tuning.

Gap 2: Uniform 80-per-intent distribution does not reflect real EEU/AAWSA traffic skew (likely heavily weighted toward top-up and outage queries).

Recommendation: Build a frequency-weighted evaluation set sampled in proportion to observed EEU/AAWSA chatbot or call-center intent distribution; report per-intent metrics alongside macro-averaged accuracy to surface deployment-relevant performance.